Test Certificate as per IEC 60034-1: 2010 & IS 12615: 2011

Certificate No. 2015I019 K-2015013 Date: 20-Mar-18

Customer Details: Test Date: 6-Jan-16

Test Results

Resistance Measurement of Stator winding

Stator Winding	Terminal	Resistance (Ω)	Temperatue (°C)
Cold	U1 – V1	9.16100	30.1
	V1-W1	9.16300	
	U1-W1	9.15700	
Hot	U1 – V1	10.62800	30.2

Resistance measurement of accessories

RTD(PT100) (Ω)					
BTD (Ω)					
PTC (Ω)					
ACH (Ω)			Other		

Temperature Rise Test(by Resistance method)

Voltage (V)	Current (A)	Power (kW)	Power Factor	Slip (%)	Coolant Temp. (°C)	Temperature Rise(°C) RTD	DE Bearing	NDE Bearing	Frame
415.8	3.22	1.829	0.79	4.904	30.2		20.9		24.7

Temperature rise after 2.45 hours 42.4 °C

Load Test and Efficiency Determination (by Indirect Method)

Output power (%)	Frequency (Hz)	Voltage (V)	Line Current (A)	Input Power (kW)	Slip (%)	Speed (RPM)	Output Power (kW)	Power Factor	Efficiency(η) %
150.0	49.97	415.30	4.74	2.934	8.580	1370.5	2.263	0.86	77.13
125.0	49.97	415.30	3.93	2.360	6.655	1399.3	1.893	0.83	80.19
100.0	49.97	415.40	3.22	1.831	4.962	1424.7	1.511	0.79	82.50
75.0	49.97	415.20	2.65	1.362	3.541	1446.0	1.138	0.71	83.57
50.0	49.97	415.20	2.19	0.921	2.269	1465.1	0.761	0.58	82.58
25.0	49.97	415.30	1.89	0.514	1.129	1482.2	0.387	0.38	75.38

Locked Rotor Test at reduced voltage and Locked rotor torque test

Frequency(Hz)	Voltage(V)	Current(A)	Power(kW)	Power Factor	Torque(Nm)	Tst/Tn	Ist/In
50.00	96.40	3.20	0.29	0.55	-	-	-
50.00	415.00	20.90	-	-	29.80	2.98	6.53

Reduced Voltage Running up test

Motor runs at full speed in specified direction on no load at Voltage (V) 111

No Load Test

Freq(Hz)	Voltage(V)	Current(A)	Power(kW)	pf	Speed
50.00	415	1.72	0.11	0.09	1499.8

Overspeed Test for 2 mins

Freq(Hz)	Speed(RPM)	Withstood
60.00	1798.00	

Shaft Voltage Test

Test Voltage(V)	Current(A)	Shaft Voltage(mV)
-	-	-

High Voltage Test

Voltage (kV)	Time(s)	Withstood
1.5	60	

Insulation Resistance Test

Insulation Resistance(MΩ)	
Before HV	>1000Mohm
After HV	>1000Mohm

Vibration severity Test at no load

Vibration (mm/s)	Axial	Horizontal	Vertical
DE	1.0	1.1	1
NDE		1	1

Polarisation Index Test

	Polarisation Index
IR_{10min}/IR_{1min}	-

Noise Level at No load

Background(dB)	Sound Level(dB)
46.8	53.8

Momentary Excess Torque test

Overload Test(%)	Time(s)	
160.00	15	Withstood

Legends: Tsn/Tn- Ratio of Starting Torque to Rated Torque of Machine

Ist/In-Ratio of Starting Current to Rated Current of Machine


CREATIVE POWER
CONTROL